QP Code: ..... Reg. No.: .....

Name: .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) EXAMINATION OCTOBER 2026

FIFTH SEMESTER

ELECTIVE – MG5DSEBTG301- BASIC BIOINFORMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S)

Students should attempt at least one question from each Course Outcome to enhance overall outcome attainability.

Part A – One Word Answer Questions

Answer any 10 questions.
Each question carries 2 marks.

1. What computational discipline deals with the storage, retrieval and analysis of biological information? [U / CO1]
2. Which repository provides access to experimentally determined three-dimensional structures of proteins and nucleic acids? [K / CO2]
3. Identify the nucleotide sequence repository maintained by NCBI. [K / CO3]
4. Recognize the program is commonly used for comparative protein structure modelling. [K/ CO3]
5. Name any one Phylogenetic Tree Construction tool. [U / CO4]
6. Recall the BLAST program designed to compare a nucleotide query sequence against a nucleotide sequence database. [K / CO3]
7. Write the term that denotes a continuous stretch of DNA that can potentially encode a protein. [K / CO3]
8. State the system used to determine the geographic position of an object on Earth. [K / CO4]

9. Mention the branch of genomics that deals with the influence of genetic variation on drug response. [K / CO4]

10. What term refers to the principles governing responsible conduct in bioinformatics? [K / CO4]

11. Write the full form of the term QSAR. [K / CO4]

12. What term refers to the principles governing responsible conduct in bioinformatics? [K / CO3]

(10 × 2 = 20 marks)

Part B – Short Essay Questions

Answer any 4 questions.
Each question carries 5 marks.

13. Summarize the basic workflow of Translation of an Unknown DNA Sequence. [U / CO2]

14. Describe the major functions of Homology Modelling and Related Bioinformatics Tools. [U / CO3]

15. Compare global and local sequence comparison methods with respect to their principles and applications. [An / CO3]

16. Analyse the differences among BLASTn, BLASTp and BLASTx and indicate a suitable application for each. [An / CO3]

17. Illustrate the major steps involved in pharmacophore-based drug design. [A / CO4]

18. Explain the applications of bioinformatics in basic biological research. [U / CO4]

(4 × 5 = 20 marks)

Part C – Essay Questions

Answer any 2 questions.
Each question carries 15 marks.

19. Explain the major goals and scientific contributions of the Human Genome Project. [U / CO1]

20. Demonstrate the biological databases into genomic, nucleotide sequence and protein databases with suitable examples. [A / CO2]

21. Design how dot matrix methods can be used to identify similarities and differences between biological sequences. [S/ CO3]

22.Analyse the major stages involved in molecular docking, from ligand selection and active-site identification to docking and result evaluation. [An / CO4]

(2 × 15 = 30 marks)